

Rijul Sudesh

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EDUCATION

Penn State University

Bachelor of Science in Software Engineering

World Campus, PA

Jan. 2022 – May 2026

- GPA 3.63/4.0 — Dean's Honor List
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Databases, Software Design

EXPERIENCE

Software Engineering Intern

Hopscotch

Mar. 2025 – May 2025

New York, NY

- Increased unit test coverage by 30%, reducing untested legacy code written in GoLang in website back-end and improving release confidence
- Troubleshoot inconsistencies within different Operating Systems using Docker containerization and DevOps workflows
- Executed 50+ manual test cases across regression and functional test suites, identifying 8 bugs before production release
- Identified and fixed recurring defects in legacy modules, contributing to a cleaner and more maintainable codebase using CI/CD pipelines

PROJECTS

NASA Psyche Card Game | C#, Unity

June 2025 – May 2026

- Developed the back-end system using C# and using Unity OOP to manage game logic and user interactions across the application
- Built highly responsive UX components within the Unity Editor for a web-based educational platform deployed on the NASA Psyche Web Portal to demonstrate mission concepts
- Conducted integration testing and resolved runtime bugs within the Unity application to ensure stable performance across target browser environments
- Coordinated with a cross-functional team including a project lead and UI designers to deliver a functional product demo to NASA Psyche Mission stakeholders

Football Victory Prediction Model | Python, Spyder, Excel, PyTorch

Aug. 2024 – Dec. 2024

- Built predictive models using PyTorch and K-means clustering to forecast NFL game outcomes from 50 years of historical data sourced from *SportsDB*
- Cleaned and processed 30,000+ records using SQL and Python, structuring raw data into a training-ready dataset
- Improved data pipeline reliability through modular code design and structured version control practices on Git

Langchain RAG Chatbot | Python, Poetry, JavaScript

Jan. 2026 – Mar. 2026

- Built a lightweight AI chatbot enabling users to upload documents and query them via a retrieval-augmented generation (RAG) pipeline, returning grounded answers using agentic workflows
- Implemented document parsing and context retrieval using LangChain, with backend APIs handling file ingestion and query processing

Fullstack Inventory Management System | Java, SQL

June 2024 – Aug. 2024

- Built a full-stack web application order and inventory DBMS project in Java for a fictional autoparts distributor
- Implemented back-end query retrieval systems and context retrieval using Oracle SQL and handling query processing

Desktop Companion | Godot, C++, Blender, Krita

May 2025 – Present

- Developed an interactive desktop companion application in Godot featuring real-time user interaction and event-driven behaviour
- Designed and integrated custom 3D assets from Blender into the Godot engine pipeline, managing the full workflow from modelling to in-engine implementation

TECHNICAL SKILLS

Languages: Python, Go, C/C++, C#, Java, SQL (Oracle), JavaScript, MIPS Assembly

Developer Tools: Git, Docker, Microsoft Azure, VS Code, Visual Studio, IntelliJ, Spyder, Blender, Krita, Figma

Libraries & Frameworks: Pandas, NumPy, Matplotlib, PyTorch, .NET